Department of Mathematical and Physical Sciences
Fall-2024; MID Term Exam
Course Name: Engineering Chemistry -1 (CHE 107)
Sec-2; Time: 70 min; Marks: 25; Date: 19/12/24
Course Instructor: Dr. Zubair Hasan

Please answer all the questions

1(a)	What were the major shortcomings of Rutherford's atomic model.	**2**
(b)	An electron in a hydrogen atom transitions from the initial state to the n=2 state, emitting a photon with a wavelength of 486 nm. Calculate the initial quantum number for this transition. ($h=6.626\times10^{-34}$ J, $c=3.0\times10^{8}$ m/s, $R_H=2.18\times10^{-18}$ J)	**3**
2(a)	Define conjugate acid base with example.	**2**
(b)	Describe the bonding in XeF_2 according to valence bond theory. Assume that the molecular geometry is the same as the VSEPR model's.	**3**
3(a)	Describe the geometry of the following compounds according to VSEPR theory: CH_4, SF_6	**2**
(b)	A solution contains benzoic acid (C_6H_5COOH, $K_a=6.3\times10^{-5}$) at a concentration of 0.9 M and hydrocyanic acid (HCN, $K_a=6.2\times10^{-10}$) at a concentration of 2.5 M. Calculate the pH of the solution. Determine the equilibrium concentration of CN^-	**3**
4 (a)	Define buffer solution. Write down the mechanism of the basic buffer.	**2**
(b)	A buffer solution is prepared using CH_3COOH and CH_3COONa. The initial concentrations of CH_3COOH and CH_3COONa are 0.3 M and 0.4 M, respectively. The acid dissociation constant for CH_3COOH□ is Ka $=1.75\times10^{-5}$. After adding 0.015 M HCl, calculate the pH of the buffer solution.	**3**
5 (a)	Define redox reaction.	**1**
(b)	Balance the following redox reaction using the ion-electron method, showing all the necessary steps:	**4**

$$MnO_4^- + I^- \rightarrow Mn^{2+} + I_2$$